

→ $L(\theta)$ = likelihood function.

→ θ = Model parameter.

→ $p(x_i; \theta)$ = Probability of observing sample x_i .

Negative Log-Likelihood.

$$NLL = -\log L(\theta).$$

→ Minimizing NLL is mathematically

Ex:

Correct class probability = 0.95

$$NLL = -\log(0.95) = 0.051$$

∴ correct class probability = 0.05.

Concept	Description	Example.
MLE	Estimates model parameters by maximizing likelihood of observed data.	Neural Network training.
Likelihood function	Measures how likely the observed data under set of parameters	Probability correct prediction
Log-Likelihood	Logarithm of likelihood function for easier optimization.	Summing likelihoods

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Advantages

→ stable convergence.

→ Accurate gradient calculation.

→ Slow for large dataset.

→ requires more memory.

2. Stochastic Gr.D

The model updates its weight after processing each individual training sample, rather than waiting for entire dataset.

3. Mini-batch Gr.D

Instead of using one sample on the entire dataset, Mini-batch Gr.D uses a small batch of samples.

Ex:

Dataset: 1000 images

Batch size: 100 images

weight updates:

100 images → Update

Next 100 images → update

Next 100 images → update.

Advantages

③ $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ multiply = $(1 \times 0) + (0 \times 1) + (2 \times 1) + (1 \times 0)$
 $= 0 + 0 + 2 + 0$
 $= 2.$

④ $\begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}$ multiply = $(0 \times 0) + (2 \times 1) + (1 \times 1) + (3 \times 0)$
 $= 0 + 2 + 1 + 0$
 $= 3.$

final feature map = $\begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$

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Unsupervised Learning

Algorithm learns patterns & structures from unlabeled data.

→ There is no predefined output.

k-Means Clustering

Unsupervised learning algorithm that divides a dataset into k clusters based on similarity b/w data points.

→ Each cluster is represented by a centroid.

* k-Means commonly used Euclidean distance.

$A(x_1, y_1)$, $B(x_2, y_2)$

distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Problems

c) Recurrent Neural Network.

- process sequential data.
- maintains memory of previous inputs.
- used in language translation & speech recognition.

d) Deep Neural Network

- contains multiple hidden layers
- can learn highly complex representations.
- used in deep learning applications.

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